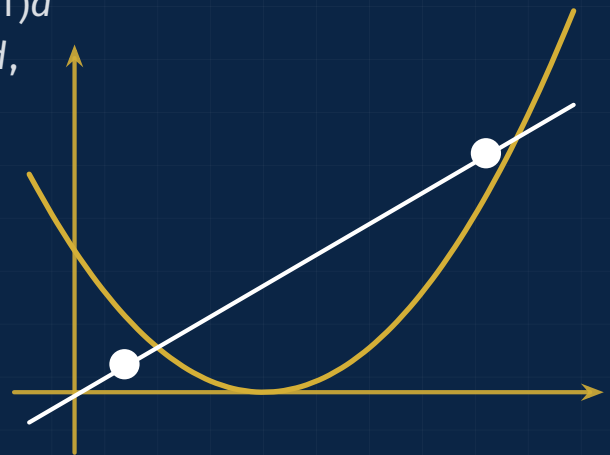


DR ESLAM AHMED
eliteigcse.com

CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 1

Sequences

Two formulas do everything: $u_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$. Name a and d , write what you know, solve.

**6**

VARIANTS

28

PAST-PAPER PARTS

30

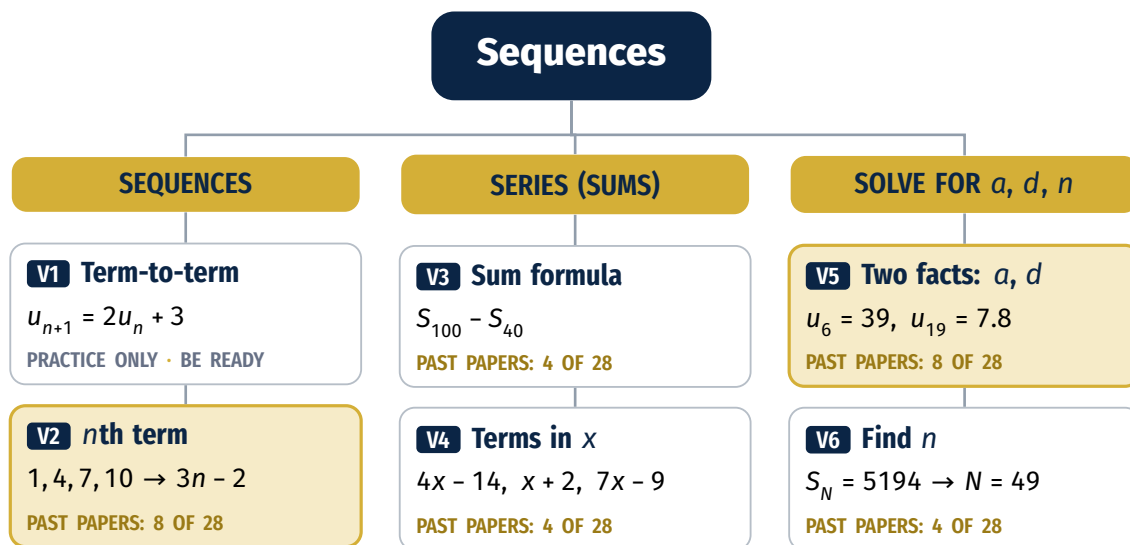
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Term-to-term **V2** n th term **V3** Sum formula**V4** Terms in x **V5** Two facts: a and d **V6** Find n

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Sequences at a glance



Your notes


★ How often does each variant come up?

25 past-paper questions (Jan 2020 → Nov 2025) hold 28 parts; 30 practice parts — grouped by type.

	PAST PAPERS (OUT OF 28)	PRACTICE PARTS
V1 Term-to-term	0	9
V2 n th term	8	14
V3 Sum formula	4	2
V4 Terms in x	4	1
V5 Two facts: a and d	8	4
V6 Find n	4	0

WHAT THIS TELLS YOU Two very different questions: an easy n th term at Q1–4 (2–3 marks) and a hard arithmetic-series question near the end (4–6 marks) on almost every paper. Both use the same two formulas.

BEFORE YOU START

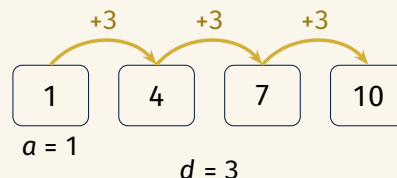
The big idea



Arithmetic = add the same d every time.

First term a , common difference d . To reach term n you add d exactly $n - 1$ times.

A **series** is the terms added up: S_n .



The two formulas (given on the formula sheet: the sum)

n th term

$$u_n = a + (n - 1)d$$

Term 20: $a + 19d$.

Sum of n terms

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

Also $S_n = \frac{n}{2}(a + \ell)$, ℓ = last term.

Words → formulas

“the 8th term is 45”

$$a + 7d = 45$$

One equation.

“sum of first 30 is 2880”

$$15(2a + 29d) = 2880$$

One equation.

“from 41st to 100th”

$$S_{100} - S_{40}$$

Take off the first 40.

“the last term is x ”

$$x = a + (n - 1)d$$

Find n first.



CALCULATOR FIRST, THEN CHECK

Check a sum on the calculator with $\sum_{n=1}^{30} (9 + 6(n - 1)) = 2880$ ✓. The number of terms n must be a positive whole number.

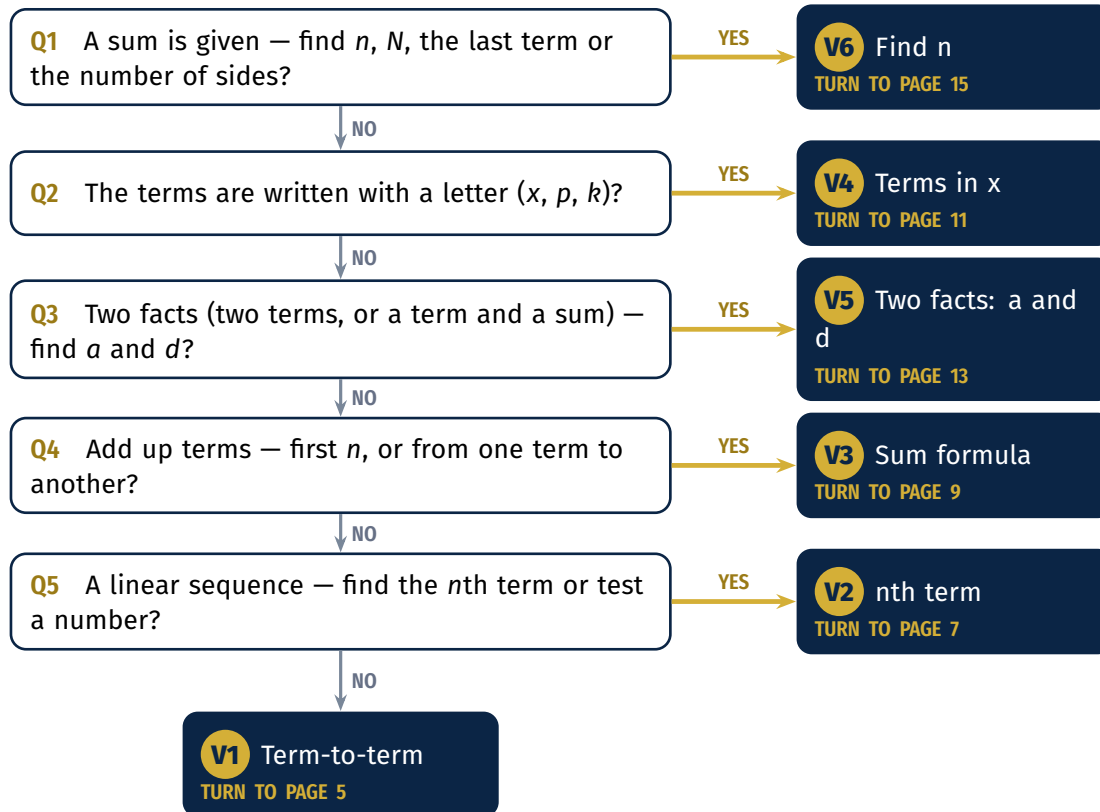
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Term-to-Term & Special Sequences

Follow the rule one step at a time; know the special sequences by sight.

SEEN IN PAST PAPERS

0 / 28

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 9



Your notes

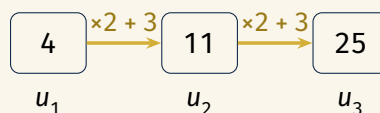
? What is it?

In plain words

A **term-to-term** rule makes the next term from the one before:

$$u_{n+1} = 2u_n + 3.$$

Know the special ones: **squares** 1, 4, 9, 16; **cubes** 1, 8, 27; **triangular** 1, 3, 6, 10; **Fibonacci-type**: add the two before.



IT LOOKS LIKE

$$u_{n+1} = 2u_n + 3$$

2	5	10	17
3	4	7	11

How do I do it?

1 Read the rule

u_{n+1} = next, u_n = this one.

2 One step

Put the term you know into the rule.

3 Repeat

Use each new term for the next.

4 Spot

Squares, cubes, triangular, Fibonacci-type.

ONE STEP AT A TIME

Write u_2 before u_3 .

V1 Term-to-Term & Special Sequences

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

" $u_{n+1} = 2u_n + 3$ and $u_1 = 4$ "

→ $u_2 = 2(4) + 3$, then u_3

"Each term is the sum of the two before"

→ **Fibonacci-type: add**

Differences 3, 5, 7 (going up by 2)

→ **Square numbers are hiding: $n^2 + \dots$**

WORKED EXAMPLE DR ESLAM

Term-to-term · 2 marks

$u_{n+1} = 2u_n + 3$ and $u_1 = 4$. Find u_3 .

STEP 1

$$u_2 = 2u_1 + 3$$

◀ the rule with $n = 1$

STEP 2

$$u_2 = 2(4) + 3 = 11$$

◀ one step

STEP 3

$$u_3 = 2(11) + 3 = 25$$

◀ again

ANSWER $u_3 = 25$

✓ $4 \rightarrow 11 \rightarrow 25$ ✓



WATCH OUT

✗ $u_3 = 2(3) + 3$

✓ $u_3 = 2u_2 + 3$

The rule uses the previous term, not n .



EXAM TIP

Write each term on its own line — a slip in u_2 still earns the method mark.



Check yourself

calculator first · answers on page 17

1.1

Dr Eslam

Each term is the sum of the two before: 3, 4, 7, 11, ... Find the next two terms.

Answer:

1.2

Dr Eslam

Find the next term of 2, 5, 10, 17, ...

Answer:

1.3

Practice · Medium

$$u_n = \frac{n}{2n-1}. \text{ Find } u_3.$$

Answer:

V2

VARIANT 2 MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

8 / 28

PRACTICE QUESTIONS: 14

The n th Term (Linear)

Difference first (dn), then fix the zero term.

Came up in: May 2021 P1H Q2 • Jun 2022 P1H Q1 • Jun 2022 P1HR Q2 • Jun 2023 P2HR Q4 • May 2024 P1H Q1

 Your notes

? What is it?

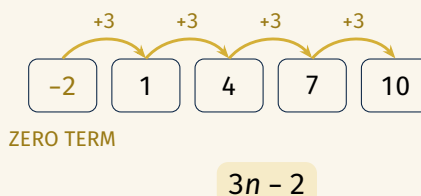
In plain words

The **difference** tells you the number in front of n . Then ask: what is the term **before** the first one (the zero term)? That is the number on the end.

Going down? The difference is negative: $91 - 6n$.

IT LOOKS LIKE

1 4 7 10 85 79 73
67 is 150 a term?



How do I do it?

1 Difference

Same gap $d \rightarrow dn$.

2 Zero term

Step back once from the first term.

3 Write

 $dn + \text{zero term}$.

4 Test

Put $n = 1$: you must get the first term.**DIFFERENCE, THEN ZERO TERM** Is 150 a term? Solve; n must be whole.

V2 The n th Term (Linear)

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Find an expression, in terms of n , for the n th term” → **Difference, zero term**

“Is 150 a term of the sequence?” → **Solve $dn + c = 150$: whole n ?**

“The $(2m)$ th term of $3n + 5$ ” → **Replace n by $2m$: $6m + 5$**

**WORKED EXAMPLE** **PAST PAPER**

May 2024 · Paper 1H · Q1 · 3 marks

(a) 1, 4, 7, 10, ... Find an expression for the n th term. (b) The n th term of another sequence is $5n + 17$. Find its 12th term.

STEP 1 $d = +3 \Rightarrow 3n$ ◀ difference

STEP 2 $1 - 3 = -2$ ◀ zero term

STEP 3 $3n - 2$ ◀ (a)

STEP 4 $5(12) + 17 = 77$ ◀ (b) put $n = 12$

ANSWER (a) $3n - 2$ (b) 77

✓ $n = 1$: $3 - 2 = 1$ ✓

**WATCH OUT**

✗ $n + 3$

✓ $3n - 2$

The difference goes in front of n , not on the end.

**EXAM TIP**

Always test $n = 1$ and $n = 2$ — it takes five seconds.

**Check yourself**

calculator first · answers on page 17

2.1

May 2021 P1H

Find the n th term of 85, 79, 73, 67, ...

Answer:

2.2

Jun 2023 P2HR

Find the n th term of 38, 31, 24, 17, ...

Answer:

2.3

Jun 2022 P1H

The n th term is $3n + 5$. Find the $(2m)$ th term.

Answer:

V3**VARIANT 3**

Using the Sum Formula

Put a, d, n into S_n ; for a middle block subtract two sums.

SEEN IN PAST PAPERS

4 / 28

PRACTICE QUESTIONS: 2


 Your notes

Came up in: Jan 2020 P2HR Q22 • Jan 2020 P2H Q25 • Oct/Nov 2020 P1H Q24 • Jan 2022 P1HR Q16

? What is it?

In plain words

$S_n = \frac{n}{2}[2a + (n-1)d]$ links four things: S_n, n, a, d . Know three, find the fourth.

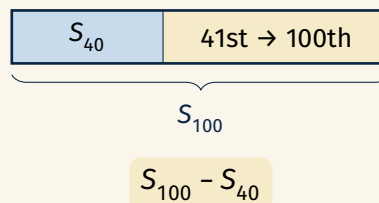
“From the 41st to the 100th term”:

all 100 minus the first 40.

IT LOOKS LIKE

$$S_{30} = 2865$$

41st to 100th



How do I do it?

1 List

$a = \dots, d = \dots, n = \dots, S_n = \dots$

2 Block?

From term p to term q : $S_q - S_{p-1}$.

3 Substitute

Into $S_n = \frac{n}{2}[2a + (n-1)d]$.

4 Solve

Find the unknown; answer the question asked.

LIST a, d, n, S FIRST The formula is on the formula sheet.

V3 Using the Sum Formula

VARIANT 3 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"The sum of all terms from the 41st to the 100th" → $S_{100} - S_{40}$

"from 2021 to 2070" → $n = 50$ (count both ends)

"first term -6, sum of first 30 is 2865" → **One unknown (d): solve**

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 1HR • Q16 • 4 marks

An arithmetic series has first term 1 and common difference 4. Find the sum of all terms from the 41st term to the 100th term inclusive.

STEP 1 $a = 1, d = 4; S_{100} - S_{40}$ ◀ list; take off the first 40

STEP 2 $S_{100} = 50(2 + 99 \times 4) = 19\,900$ ◀ substitute

STEP 3 $S_{40} = 20(2 + 39 \times 4) = 3160$ ◀ substitute

STEP 4 $19\,900 - 3160 = 16\,740$ ◀ subtract

ANSWER 16 740✓ 60 terms, average $\frac{161+397}{2} = 279$: 60×279 ✓**WATCH OUT**✗ $S_{100} - S_{41}$ ✓ $S_{100} - S_{40}$

The 41st term must stay in.

EXAM TIP

Years 2021 to 2070: $2070 - 2021 + 1 = 50$
— count both ends.

Check yourself

calculator first • answers on page 17

3.1 Oct/Nov 2020 P1H

8, 15, 22, 29, ... Find the sum of all terms from the 50th to the 100th inclusive.

Answer:

3.2 Jan 2020 P2HR

First term -6; the sum of the first 30 terms is 2865. Find the 9th term.

Answer:

3.3 Jan 2020 P2H

Saves \$50 in 2021, then k more each year to 2070. Total \$33 125. Find k .

Answer:

V4

VARIANT 4

Terms Written in x

Arithmetic means equal gaps: second – first = third – second.

SEEN IN PAST PAPERS

4 / 28

PRACTICE QUESTIONS: 1

 Your notes

Came up in: Nov 2021 P2H Q20 • Jun 2023 P2HR Q23 • May 2024 P1HR Q23 • Jun 2025 P1HR Q24

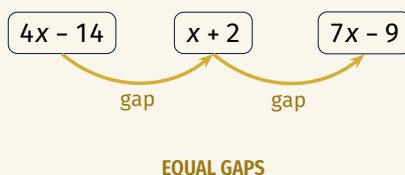
? What is it?**In plain words**

Three terms with a letter: $4x - 14$, $x + 2$, $7x - 9$. Arithmetic means the

gaps are equal, so

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2).$$

Solve for x , find the real terms, a and d — then it is V3 or V6.

**IT LOOKS LIKE** $8p$ $7p - 3$ $4p + 2$ **How do I do it?****1 Equal gaps**

$$2\text{nd} - 1\text{st} = 3\text{rd} - 2\text{nd}.$$

2 Solve

Linear equation for x .

3 Real terms

Substitute: a and d as numbers.

4 Finish

Sum (V3) or find n (V6).

EQUAL GAPS Then it is an ordinary series.

V4 Terms Written in x

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

"The first three terms are $(4x - 14)$, $(x + 2)$, $(7x - 9)$ "

→ Equal gaps → x

"Find, as an integer, the sum of the first 40 terms"

→ Real a, d , then S_{40}

Terms $k, \frac{3k}{4}, \frac{k}{2}, \dots$

→ $d = -\frac{k}{4}$ WORKED EXAMPLE **PAST PAPER**

May 2024 · Paper 1HR · Q23 · 4 marks

The first three terms of an arithmetic sequence are $(4x - 14)$, $(x + 2)$, $(7x - 9)$. Find, as an integer, the sum of the first 40 terms.

STEP 1 $(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$ ◀ equal gaps

STEP 2 $-3x + 16 = 6x - 11 \Rightarrow x = 3$ ◀ solve

STEP 3 $-2, 5, 12 : a = -2, d = 7$ ◀ real terms

STEP 4 $S_{40} = 20(-4 + 39 \times 7) = 5380$ ◀ sum

ANSWER 5380

✓ gaps 7 and 7 ✓

 WATCH OUT

$$\times (x + 2) - (4x - 14) = -3x - 12$$

$$\checkmark = -3x + 16$$

Subtracting a bracket flips both signs.

 EXAM TIP

Write the three real terms — they prove x is right.

 Check yourself

calculator first · answers on page 17

4.1 Jun 2023 P2HR

First three terms $8p$, $7p - 3$, $4p + 2$. The sum of the first n terms is -1914 . Find n .

Answer:

4.2 Nov 2021 P2H

Terms $k, \frac{3k}{4}, \frac{k}{2}, \frac{k}{4}$. The 15th term is $90 + 2k$. Find the sum of the first 30 terms.

Answer:

4.3 Jun 2025 P1HR

Terms $(2x + 5)$, $(3y - 4)$, $(4x - 2)$. The sum of the first 9 terms is 216. Find x and y .

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

8 / 28

PRACTICE QUESTIONS: 4

Two Facts: Find a and d

Each fact is one equation in a and d ; solve them simultaneously.

Came up in: Oct/Nov 2020 P1HR Q18 • Jan 2021 P1HR Q21 • Jan 2021 P1H Q24 • Jan 2022 P2H Q26 • Jun 2022 P1HR Q25 • May 2023 P1H Q20 • Nov 2024 P1H Q19 • Jun 2025 P1H Q24


Your notes

? What is it?

In plain words

"The 6th term is 39" $\Rightarrow a + 5d = 39$.

"The 19th term is 7.8" $\Rightarrow$

$a + 18d = 7.8$.

Two facts, two equations, two unknowns:

simultaneous equations. Subtract to kill a . Then answer what is asked.

IT LOOKS LIKE

$$u_6 = 39$$

$$u_{19} = 7.8; S_{30} = 4395$$

$$u_{10} + u_{20} = 284$$

$$a + 18d = 7.8$$

$$a + 5d = 39$$

$$\begin{array}{r} - \\ 13d = -31.2 \end{array}$$

$$d = -2.4, a = 51$$

How do I do it?

1 Fact 1

Write as an equation in a, d .

2 Fact 2

Write as an equation in a, d .

3 Solve

Subtract (or substitute) — Chapter 2 simultaneous.

4 Answer

Use a, d for the term or sum asked.

ONE FACT = ONE EQUATION 8 of the 28 past-paper parts.

V5 Two Facts: Find a and d

VARIANT 5 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"the 6th term is 39 ... the 19th term is 7.8" → $a + 5d = 39, a + 18d = 7.8$

"the sum of the first 10 terms is 4 times the sum of the first 5" → $S_{10} = 4S_5 \rightarrow d = 2a$

"Show clear algebraic working" → **Write both equations before solving**

WORKED EXAMPLE PAST PAPER

Oct/Nov 2020 · Paper 1HR · Q18 · 4 marks

In an arithmetic series, the 6th term is 39 and the 19th term is 7.8. Work out the sum of the first 25 terms.

STEP 1 $a + 5d = 39, a + 18d = 7.8$ ◀ two facts

STEP 3 $13d = -31.2 \Rightarrow d = -2.4, a = 51$ ◀ subtract; back-substitute

STEP 4 $S_{25} = \frac{25}{2}[102 + 24(-2.4)] = 555$ ◀ the sum

ANSWER 555
 $\checkmark u_{25} = 51 - 57.6 = -6.6; \frac{25}{2}(51 - 6.6) = 555 \checkmark$
WATCH OUT

✗ 6th term = $a + 6d$

✓ $a + 5d$

To reach term 6 you add d five times.

EXAM TIP

Check with the other form: $S_n = \frac{n}{2}(a + l)$.

Check yourself

calculator first · answers on page 17

5.1 Jun 2025 P1H

30 terms; the 20th term is 123; the sum of the 30 terms is 2880. Find a and d .

Answer:

5.2 Jun 2022 P1HR

$S_{10} = 4S_5$ and the 8th term is 45. Find the first term.

Answer:

5.3 Nov 2024 P1H

$S_{30} = 4395$; the 10th term + the 20th term = 284. Find S_{45} .

Answer:

V6

VARIANT 6

Find the Number of Terms

Put S_n equal to the total: a quadratic in n ;
keep the positive whole root.

SEEN IN PAST PAPERS

4 / 28

PRACTICE QUESTIONS: 0


Your notes

Came up in: May 2021 P1H Q23 • Nov 2023 P1H Q19 • May 2024 P1H Q24 • Nov 2025 P2H Q25

? What is it?**In plain words**

When the **sum is given** and n is unknown, $S_n = \frac{n}{2}[2a + (n - 1)d]$ becomes a **quadratic in n** .

Rearrange to $= 0$, factorise or use the formula, and keep the **positive whole** answer.

IT LOOKS LIKE

$$S_N = 5194$$

last term x

polygon angles

$$N(2N + 8) = 5194$$

$$N^2 + 4N - 2597 = 0$$

$$(N - 49)(N + 53) = 0$$

$$N = 49$$

$$N = -53 \times$$

How do I do it?**1 Find a, d**

From the facts (V4 or V5).

2 Set up

$$\frac{n}{2}[2a + (n - 1)d] = \text{the sum.}$$

3 QuadraticExpand, $= 0$, solve.**4 Choose**

Positive whole n ; answer the question (last term, angle sum ...).

QUADRATIC IN n Reject negatives and fractions.

V6 Find the Number of Terms

VARIANT 6 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"The sum of the first N terms is 5194.
Find N "

→ $S_N = 5194$: **quadratic in N**

"The last term is x ; the sum is 7260"

→ **Find n , then $x = a + (n - 1)d$**

"Interior angles form an arithmetic sequence"

→ **Sum** = $(n - 2) \times 180$

WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 2H · Q25 · 5 marks

The first term of an arithmetic series is 10. The 20th term is 86. The sum of the first N terms is 5194. Work out the value of N .

STEP 1 $10 + 19d = 86 \Rightarrow d = 4$ ◀ find d

STEP 2 $\frac{N}{2}[20 + 4(N - 1)] = 5194$ ◀ set up

STEP 3 $2N^2 + 8N = 5194 \Rightarrow N^2 + 4N - 2597 = 0$ ◀ quadratic

STEP 4 $(N - 49)(N + 53) = 0 \Rightarrow N = 49$ ◀ positive root

ANSWER $N = 49$

✓ $\frac{49}{2}(20 + 192) = 5194$ ✓

WATCH OUT

✗ keeping $N = -53$

✓ $N = 49$ only

A number of terms is positive and whole.

EXAM TIP

Big numbers? Use the calculator's equation solver, but write the quadratic first.

Check yourself

calculator first · answers on page 17

6.1 Nov 2023 P1H

3, 7, 11, 15, ... The last term is x ; the sum of the terms is 7260. Find x .

Answer:

6.2 May 2021 P1H

The sum of the first N terms is 292; the 2nd term is 8.5; the 5th term is 13. Find N .

Answer:

6.3 May 2024 P1H

A polygon's interior angles form an arithmetic sequence: smallest 84° , $d = 4^\circ$. Find the angle sum.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 18, 29

1.2 26

1.3 $\frac{3}{5}$

2.1 $91 - 6n$

2.2 $45 - 7n$

2.3 $6m + 5$

3.1 26 826

3.2 $u_9 = 50$

3.3 $k = 25$

4.1 $n = 29$

4.2 1575

4.3 $x = 5.5, y = \frac{22}{3}$

5.1 $a = 9, d = 6$

5.2 $a = 3$

5.3 9630

6.1 $x = 239$

6.2 $N = 16$

6.3 7740°

THE WHOLE TOPIC ON ONE PAGE

Sequences cheat sheet

**V1 Term-to-term**

$$u_{n+1} = 2u_n + 3$$

One step at a time.

V2 nth term 8/28

$$3n - 2$$

Difference, then zero term.

V3 Sum formula 4/28

$$S_{100} - S_{40}$$

List a, d, n ; subtract for a block.

V4 Terms in x 4/28

gap = gap

Solve for x , then real terms.

V5 Two facts 8/28

$$a + 5d = 39$$

One fact, one equation; simultaneous.

V6 Find n 4/28

$$N^2 + 4N - 2597 = 0$$

Quadratic; positive whole n .



TOP 3 LOST MARKS 1. $u_6 = a + 6d \times$ 2. $S_{100} - S_{41}$ for "41st to 100th" $\times$ 3. keeping a negative $n \times$

BOTTOM LINE $u_n = a + (n - 1)d \cdot S_n = \frac{n}{2}[2a + (n - 1)d] \cdot$ one fact, one equation $\cdot$
 n is a positive whole number.